Introduction

This page describes the **modeling guidelines for IT Products** in the context of enterprise architecture at Porsche. It is based on the general modeling guidelines of PAG and includes specific instructions for creating and maintaining IT Products in LeanIX. The creation and editing of an IT Product fact sheet follows the standard procedure for creating a general fact sheet as described in "How to create a fact sheet".

Decision Guidelines: IT Product vs. IT Component in LeanIX

Core Decision Rule

To model your IT landscape effectively in LeanIX, it is essential to distinguish between **IT Products** and **IT Components**. Use the following rule:

If the artifact...	**Then model it as...**
...directly supports a business process	**IT Product**
...implements a Business Capability	**IT Product**
...implements a Technical Capability	**IT Component**

This rule is typically sufficient for accurate classification.

Characteristics of an IT Product

An **IT Product** is defined by the following attributes:

- It bundles business functions in a **logically and functionally bounded** way to process data.
- It may include **custom-developed elements** such as application code, configuration data, macros, etc.
- It can be a **third-party software product**, used as SaaS, cloud-based, or on-premises.
- It may represent a **bundle of technical components** that together provide infrastructure or business platforms.

Examples

Name	Description	Classification	Justification
Thunder	Web application supporting the sales process	IT Product	Supports multiple business capabilities like "Vehicle Sales"
Python	Programming language	IT Component	Not directly tied to a business capability
Paint.NET	Image editing software	Neither	No direct business relevance in the organization
Oracle DB 19c	Database system	IT Component	Supports data storage, which is a technical capability
LeanIX	SaaS for Enterprise Architecture	IT Product & IT Component	Application: supports business capabilities; Component: software product
Group Retail Portal	Platform for retailer access	IT Product	Supports business and IT security capabilities
Excel	Spreadsheet software	IT Component	Not tied to a specific business process
Adobe Reader DC	PDF viewer and signer	IT Component	Signing is a technical skill, not a business capability
CATIA	3D design software	IT Product (multiple)	Used in different business contexts with distinct responsibilities

---

Modeling Rule: When to Create a Separate IT Product

Even if the same software is used across departments (e.g., CATIA, MATLAB), a **separate IT Product** must be modeled if:

- It is used in a **different business or logical context**
- It processes **different types of data**, possibly with different classifications
- It involves **different organizational responsibilities**
- It runs on **different infrastructures** (e.g., cloud vs. on-prem, Windows vs. Linux)

Structure Summary

Section	Key Concept	Description	Example(s)
Decision Rule	Classification Logic	Defines when to use IT Product vs. IT Component	Business Capability → IT Product; Technical Capability → IT Component
IT Product Characteristics	Definition	Logical bundle of business functions, may include custom or third-party software	SaaS, on-prem apps, business platforms
Examples	Real-world Mapping	Classification of common tools and systems	Thunder = IT Product; Python = IT Component
Modeling Rule	Contextual Differentiation	When to model separate IT Products for the same software	CATIA used in different departments
Responsibilities	Stakeholder Guidance	Ensures correct modeling by business and IT roles	Business units define context-specific IT Products

---

General Modeling Rules

Subscriptions

In LeanIX, **subscriptions** are used to assign individuals and roles to a Fact Sheet. There are two main types of subscriptions:

- **Responsible**: Individuals accountable for the content and maintenance of the Fact Sheet.
- **Observer**: Individuals who are informed but not responsible for the content.

The specific roles that can be assigned to a Fact Sheet are defined in the modeling guidelines for each Fact Sheet type.

Responsibility Requirement

Each Fact Sheet must have **at least one person assigned as Responsible**. This person (or persons):

- Is accountable for the **accuracy, completeness, and currency** of the Fact Sheet.
- Receives **notifications** about changes to the Fact Sheet.

- Acts as the **primary contact** for questions, clarifications, or requirements related to the Fact Sheet.

Change of Responsibility

If a Responsible person:

- Changes roles,
- Leaves the company or department,
- Or is no longer involved with the Fact Sheet,

…it is their duty to ensure a **successor** is assigned to maintain the Fact Sheet.

Automated Quality Checks

LeanIX includes automated processes to ensure data quality. These processes include:

1. **Inactive Responsible Person**
   If a Responsible person is no longer active in the company or subsidiary (e.g., due to departure), the system checks for other Responsible persons:
   - If none exist, the **direct manager** of the former Responsible is identified and automatically assigned as the new Responsible.
   - The manager also receives a **To-Do** to assign a new, appropriate Responsible person.
   - This escalation continues up to the **department head level** if necessary.
2. **No Responsible Assigned**
   If no Responsible is assigned to a Fact Sheet:
   - The system uses the **"Responsible IT Department"** relation to identify the appropriate manager.
   - This manager is then assigned as the Responsible person and is expected to **nominate a qualified business owner**.

Lifecycle Management in LeanIX Fact Sheets

Each **Fact Sheet type** in LeanIX can be assigned a **lifecycle**. Whether this assignment is **mandatory** depends on the specific Fact Sheet type and is indicated on its detail page.

Recommended Practice

Regardless of whether the lifecycle is mandatory, it is **strongly recommended** to provide the following information:

- **Start of productive use** (Go-Live date)

- **Decommissioning date** (if known)

Depending on the desired level of detail, additional lifecycle milestones can also be modeled (e.g., development start, pilot phase, sunset planning).

Benefits

Accurate lifecycle data supports:

- **Strategic planning** and **portfolio management**
- **Compliance** with regulatory and audit requirements
- **Impact analysis** for changes and decommissioning
- **Architecture governance** across all layers

Organizational Responsibilities

Each Fact Sheet in LeanIX must include an assignment of **organizational responsibility**. This responsibility can be defined at two levels:

- **Organizational Structure Level** (e.g., departments, business units)
- **Process Organization Level** (e.g., Agile Release Trains, portfolios)

For certain Fact Sheet types, a distinction may also be made between **functional (business) responsibility** and **technical (IT) responsibility**.

> ✅ At least one form of responsibility must be specified—either at the organizational structure level or the process organization level.
> 🔄 Ideally, both levels should be documented to ensure full traceability and accountability.
>
> ⚠️ **If IT responsibility is defined as mandatory in the modeling guidelines for a specific Fact Sheet type, it must be provided without exception.**

Responsibility at the Organizational Structure Level

At Porsche, the **organizational structure** is defined by the official organizational chart, which organizes the company into hierarchical levels such as:

- Division
- Main Department
- Department

- Subdepartment (lowest level)

  In LeanIX modeling, unless explicitly stated otherwise, responsibility must always be assigned to the **lowest available level** of the organizational structure.

Key Principles

- Always reference the **most granular organizational unit** available.
- Ensure that the assigned unit reflects the **actual operational responsibility**.
- This ensures **clear accountability**, **accurate reporting**, and **effective governance**.

Responsibility at the Process Organization Level

Many IT products and applications at Porsche are developed and operated by **Agile Release Trains (ARTs)**. Others are managed within specific **portfolios**.

Assignment Guidelines

- If an **ART** is responsible for a product (or any other relevant Fact Sheet type), this responsibility must be **explicitly documented** in LeanIX by assigning the corresponding ART Fact Sheet.
- If the responsibility lies with a **single portfolio**—as is often the case with **P51 applications**—then assigning the portfolio is **sufficient**.

  ✅ Always ensure that the assigned ART or portfolio accurately reflects the **operational ownership** of the product.

Key Principles

- Use the **ART Fact Sheet** when agile teams are responsible for development and operations.
- Use the **Portfolio Fact Sheet** when responsibility is anchored in a business or IT portfolio.
- This ensures **clear accountability**, supports **agile governance**, and aligns with **LeanIX modeling standards**.

Please note

Only organizational fact sheets that are tagged with "**Synced**" or "**Manual**" may be used! The use of fact sheets with the tag "**Legacy**" is not allowed!

Modeling Version History with Successor and Predecessor Relations

In LeanIX, the **Successor** and **Predecessor** relations are used to model the **version history** of Fact Sheets over time.

These relations should be applied whenever a **new version** of an existing Fact Sheet is created to **replace** an older version.

> ✅ Successor relations can also be used **while the previous version is still active** and in productive use.

Best Practice for Unused Relations

If no predecessor or successor exists for a given Fact Sheet, it is recommended to **explicitly set the relation field to**:

> **"Leave empty"**

This ensures clarity in the data model and avoids ambiguity in version tracking.

Key Principles

- Use **Successor/Predecessor** to document version transitions.
- Apply even if the older version is still operational.
- If not applicable, set the relation field to **"Leave empty"** instead of leaving it undefined.

Modeling Hierarchies with Parent/Child Relations in LeanIX

In LeanIX, **Parent/Child relations** are used to model **hierarchical dependencies** between Fact Sheets. These relations help structure complex systems, portfolios, or organizational units by defining **logical containment** or **ownership hierarchies**.

Usage Guidelines

- A **Parent Fact Sheet** represents a higher-level entity that logically contains or governs one or more **Child Fact Sheets**.
- A **Child Fact Sheet** is a subordinate entity that inherits context or structure from its parent.

This relation is commonly used in the following scenarios:

Use Case	Parent Type	Child Type
Application Suite	Application	Application
Platform Composition	Platform	IT Component
Organizational Hierarchy	Organization Unit	Sub-Unit
Portfolio Structure	Portfolio	Product / Application

Special Rule for IT Products

> ⚠️ **For the Fact Sheet type "IT Product", the Parent/Child relation must only be used after prior consultation with the responsible Portfolio Architect and the LeanIX team.**
> This ensures that hierarchical modeling aligns with enterprise architecture principles and avoids misrepresentation of product structures.

Best Practices

- Use Parent/Child relations to **group related Fact Sheets** under a common context.
- Ensure that the hierarchy reflects **actual ownership or architectural structure**.
- Avoid circular or ambiguous relationships.
- Document the rationale for the hierarchy in the **description field** of the Fact Sheet if needed.
- ✅ **If the Parent/Child relation is not applicable**, explicitly set the field to **"Leave empty"** to ensure clarity and data quality.

Benefits

- Improves **navigability** and **traceability** in LeanIX.
- Supports **impact analysis** and **portfolio planning**.
- Enables **structured reporting** and **governance alignment**.

Specific Modelling Rules and Best Practices for an IT Product

When an IT Product is identified, it must be modeled using an **IT Product Fact Sheet** in LeanIX. The following attributes and relationships are required:

- **Category**
- **Naming**
- **Versioning**
- **Description**
- **Interfaces**
- **Business Architecture**
- **Technical Architecture**
- **Governance**
- **Enterprise Portfolio Management**

Category Assignment

Each IT Product must be assigned to one of the following categories:

Category	Description	Examples
Application	Provides functionality via graphical or text-based user interfaces	CATIA, Matlab, LeanIX, Vector CANoe
Platform	Provides technical infrastructure	CLUU, Cloud Account Management, Dataiku
Group Application	Applications developed and operated by Group brands and used by Porsche	eAkte, EXAM, FLAUDO

Naming Conventions

- Names should be **technology- and product-neutral** where possible, and reflect the **business context**.
- Use terms familiar to users (e.g., product name, vendor name, or internal nickname).
- Examples:
    - A CATIA-based application may be named **"CATIA"** or **"Lighting Simulation"**.
    - An Excel macro used in HR and known as "HR-Manager" should be named accordingly.
    - LeanIX as a product is simply named **"LeanIX"**.

🔒 Important: No two IT Products in LeanIX can have the same name and version. If a name conflict occurs:

- Check if a new version should be created.
- Otherwise, assign a new, unique name.

Versioning

- Versioning is managed via the **"Release"** attribute.
- The release refers to the **functional version** of the product, not the technical component version.
- Initial release should be set to **"1.0"**.
- A new version is required when:
  - The **technical architecture** changes (e.g., migration to cloud).
  - The **responsibility** shifts to another department or portfolio.

Link versions using the **Successor/Predecessor** relationship.

Description

- Provide a clear description of the product in its **business context**.
- Include notes on **GoBD relevance** if applicable.
- The description must be written in **English**.

Interfaces

- If the IT Product **provides or consumes interfaces**, link or create the appropriate **Interface Fact Sheets**.

Business Architecture

Define the business context by linking the IT Product to:

- **Digital Solution** (if developed under SAFe/ART)
- **Business Capabilities**
- **Processes** (via BIC-LeanIX sync only)

Digital Solution

- If the product is part of an **Agile Release Train (ART)**, link it to the corresponding **Digital Solution**.
- If the required Digital Solution is missing, request it from the responsible Program Management team.

Business Capabilities

- Link supported capabilities from the maintained **Business Capability Map**.
- Always set the **Support Type** to indicate how the product supports the capability.

Processes

- Process links are managed **automatically** via the **BIC-LeanIX synchronization**.
- Manual linking is **not allowed**.

Technical Architecture

Each IT Product must include a description of its **technical architecture**, including:

- **Software components**
- **Platforms**
- **Hosting type and provider** (mandatory)

On-Premises Architecture

Model the following components:

- Deployed software product
- OS platforms (e.g., Windows, Linux)
- Databases and DBMS platforms
- Runtime environments (e.g., Tomcat)
- Container platforms
- Required services

Ensure all components are documented in the correct version.

Cloud Architecture

- Cloud services are **automatically created** if the product runs on AWS or Azure via PAG Cloud Application Management.
- Only deployed software products need to be modeled manually.
- For SaaS products, model the **SaaS as an IT Component** and link it to the product.

Client Architecture

- For client-side software, assign the installed product as a component.
- Also assign the correct **client type** (e.g., BK-Client, Tech-Client) and platform version.

Governance

Each IT Product must include governance-related information from the following areas:

- **IT Security**
- **GoBD**
- **Export Control**
- **Document Classification**
- **P51 Assignment**
- **Data Protection**

Refer to the internal governance documentation for responsibilities and further details.

Enterprise Portfolio Management

Enterprise Architecture attributes for classification and evaluation include:

- **TIME Classification**
- **7R Classification**

  For questions, contact your Enterprise Architecture team.

Structure Summary

Section	Attribute / Concept	Description	Notes / Examples
IT Product Fact Sheet	Required Elements	Category, Naming, Versioning, Description, Interfaces, Architecture, etc.	—

Section	Attribute / Concept	Description	Notes / Examples
Category	Application / Platform / Group App	Defines the type of IT Product	CATIA = Application
Naming	Contextual, user-friendly	Should reflect business use and be unique	"HR-Manager"
Versioning	Release	Functional versioning, not technical	Start with 1.0
Description	Business context	In English, include GoBD relevance if applicable	—
Interfaces	Provided/Consumed	Link to Interface Fact Sheets	—
Business Architecture	Digital Solution, Capabilities, Processes	Defines business alignment	SAFe, ART
Technical Architecture	Components, Platforms	Includes hosting type, software, OS, DB, runtime, etc.	On-Prem, Cloud
Governance	Compliance areas	IT Security, GoBD, Export Control, etc.	—
Enterprise Portfolio Management	TIME, 7R Classification	For strategic evaluation	—

Metamodel View

The diagram shows the possible relations between the entities that can be used to model an **IT Product** in the context of the appropriate architecture level.

LLM-Ready Structure Summary

IT Product Metamodel – LeanIX Enterprise Architecture Context

Overview

The image presents a **metamodel** of an **IT Product** within an enterprise architecture framework, modeled in LeanIX. It visually defines the relationships between the IT Product and other architectural elements, aligning with **TOGAF® standards**.

Central Element

- **IT Product** (highlighted in green):
  The core entity in the model, representing a business-facing software solution or application that supports business capabilities and processes.

Connected Elements and Relationships

Element	Relationship Type	Description
Initiative	*Affected by*	The IT Product is influenced or impacted by strategic or operational initiatives.
Digital Solution	*Supports*	The IT Product is part of or supported by a Digital Solution, especially in agile (SAFe) contexts.
Organization	*Responsible for*	A business unit or team is accountable for the lifecycle and governance of the IT Product.
Interface	*Provides / Consumes*	The IT Product either offers or uses interfaces to exchange data or functionality.
Process	*Supports*	The IT Product enables or automates specific business processes.
Business Capability	*Supports*	The IT Product contributes to the realization of one or more business capabilities.
IT Component	*Requires*	The IT Product depends on technical components (e.g., software, platforms) for implementation.
ADR (Architecture Decision Record)	*Supports*	The IT Product is linked to architectural decisions that guide its design and evolution.

TOGAF® Alignment

The model aligns with the following **TOGAF® architecture domains**:

TOGAF® Layer	Description
A. Architecture Vision	High-level strategic alignment of the IT Product with business goals.
B. Business Architecture	Captures the business processes and capabilities supported by the IT Product.
C. Technical Architecture	Describes the technical components and platforms required by the IT Product.

Purpose and Usage

This metamodel serves as a **reference structure** for modeling IT Products in LeanIX. It ensures consistency, traceability, and alignment with enterprise architecture principles. It is particularly useful for:

- **Enterprise Architects**: to maintain architectural integrity and trace dependencies.
- **Business Analysts**: to understand business impact and capability support.
- **Business Units**: to manage ownership, governance, and compliance.